Examiner only

5. *(a)* Gwyn carried out an experiment to find the solubility of potassium nitrate at room temperature, 25 °C. The method he used is shown in stages **1**-**3**.

Stage **1**

He added solid potassium nitrate to 50 cm^3 of water until a saturated solution was formed.

⇩

Stage **2**

He pipetted 25 cm^3 of the saturated solution into a pre-weighed evaporating basin. He put the basin and the solution into a warm oven.

⇩

Stage **3**

He removed the basin from the oven and allowed it to cool. He weighed the basin and crystals after 2 hours.

His results are shown below.

Mass of evaporating basin	= 42.6 g
Mass of evaporating basin and potassium nitrate crystals	= 54.2 g

(i) State what is meant by a *saturated* solution. [1]

....

....

(ii) Use Gwyn's results to calculate the solubility of potassium nitrate in g / 100 g of water. [2]

solubility = ... g / 100 g of water

Examiner only

(iii) Gwyn's experimental value for the solubility of potassium nitrate is much bigger than expected. Suggest how Gwyn should adapt stage **3** of his method in order to get a more accurate value. Explain your reasoning. [3]

....

....

....

....

....

(iv) Gwyn was asked to find out if sodium chloride is more soluble in water than potassium nitrate. State which variable Gwyn **must keep the same** in his experiment in order for him to be able to compare the two solubilities. [1]

....

(b) (i) The solubility of potassium bromide at 20 °C is 64 g / 100 g of water. Calculate the mass of solid that forms when a solution containing 43.9 g of potassium bromide in 50 g of water is cooled to 20 °C. [2]

mass = ... g

(ii) Calculate the number of moles of potassium bromide, KBr, in 64 g. Give your answer to **two** significant figures. [2]

number of moles = ... mol

11